

1. Загрузка/удаление модуля

Соберём модуль ядра:

```
vadim@vadim-RV520:~/Projects/eltex-embedded-c-homework/homework-20-kernel-modules$ make
make -C /lib/modules/6.17.0-35-generic/build M=/home/vadim/Projects/eltex-embedded-c-homework/homework-20-kernel-modules modules
make[1]: Entering directory '/usr/src/linux-headers-6.17.0-35-generic'
make[2]: Entering directory '/home/vadim/Projects/eltex-embedded-c-homework/homework-20-kernel-modules'
warning: the compiler differs from the one used to build the kernel
  The kernel was built by: x86_64-linux-gnu-gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
  You are using:          gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
  CC [M]  hw20_mod.o
  MODPOST Module.symvers
WARNING: modpost: missing MODULE_DESCRIPTION() in hw20_mod.o
  CC [M]  hw20_mod.mod.o
  CC [M]  .module-common.o
  LD [M]  hw20_mod.ko
  BTF [M] hw20_mod.ko
Skipping BTF generation for hw20_mod.ko due to unavailability of vmlinux
make[2]: Leaving directory '/home/vadim/Projects/eltex-embedded-c-homework/homework-20-kernel-modules'
make[1]: Leaving directory '/usr/src/linux-headers-6.17.0-35-generic'
vadim@vadim-RV520:~/Projects/eltex-embedded-c-homework/homework-20-kernel-modules$
```

Проверим загрузку/выгрузку модуля:

```
vadim@vadim-RV520:~/Projects/eltex-embedded-c-homework/homework-20-kernel-modules$ sudo insmod hw20_mod.ko
[sudo] password for vadim:
vadim@vadim-RV520:~/Projects/eltex-embedded-c-homework/homework-20-kernel-modules$ dmesg | tail -n 2
[ 664.954365] EXT4-fs (sda5): unmounting filesystem fcecf7bf-a032-4469-92b0-cbf9ea59d4c2.
[ 789.118301] hw20_mod: Module loaded. Major number is 234.
vadim@vadim-RV520:~/Projects/eltex-embedded-c-homework/homework-20-kernel-modules$ sudo rmmod hw20_mod.ko
vadim@vadim-RV520:~/Projects/eltex-embedded-c-homework/homework-20-kernel-modules$ dmesg | tail -n 3
[ 664.954365] EXT4-fs (sda5): unmounting filesystem fcecf7bf-a032-4469-92b0-cbf9ea59d4c2.
[ 789.118301] hw20_mod: Module loaded. Major number is 234.
[ 802.457449] hw20_mod: Module unloaded.
vadim@vadim-RV520:~/Projects/eltex-embedded-c-homework/homework-20-kernel-modules$
```

2. Character device

Произведём модификацию переменной в модуле через файл устройства:

```
vadim@vadim-RV520:~/Projects/eltex-embedded-c-homework/homework-20-kernel-modules$ sudo insmod hw20_mod.ko
vadim@vadim-RV520:~/Projects/eltex-embedded-c-homework/homework-20-kernel-modules$ cd /dev
vadim@vadim-RV520:/dev$ dmesg | tail -n1
[ 877.466136] hw20_mod: Module loaded. Major number is 234.
vadim@vadim-RV520:/dev$ sudo mknod hw20_mod c 234 0
vadim@vadim-RV520:/dev$ head hw20_mod
Hello!
vadim@vadim-RV520:/dev$ echo "Privet!" | sudo tee hw20_mod > /dev/null
vadim@vadim-RV520:/dev$ head hw20_mod
Privet!
vadim@vadim-RV520:/dev$
```

3. proc

Аналогично изменим переменную через файл, находящийся в псевдо-файловой системе **proc**:

```
vadim@vadim-RV520:/dev$ cd /proc
vadim@vadim-RV520:/proc$ ls -ld hw20_mod
-rw-rw-rw- 1 root root 0 Jun 16 17:05 hw20_mod
vadim@vadim-RV520:/proc$ head hw20_mod
Privet!
vadim@vadim-RV520:/proc$ echo "ABCDEEEEEEE!" > hw20_mod
vadim@vadim-RV520:/proc$ head hw20_mod
ABCDEEEEEEE!
vadim@vadim-RV520:/proc$ head /dev/hw20_mod
ABCDEEEEEEE!
vadim@vadim-RV520:/proc$
```

4. sysfs

Напоследок произведём общение с переменной ядра через **sysfs**:

```
vadim@vadim-RV520:/proc$ cd /sys/kernel/hw20_kobj
vadim@vadim-RV520:/sys/kernel/hw20_kobj$ ls -l
total 0
-rw-r--r-- 1 root root 4096 Jun 16 17:06 mod_buff
vadim@vadim-RV520:/sys/kernel/hw20_kobj$ head mod_buff
ABCDEEEEEEE!
vadim@vadim-RV520:/sys/kernel/hw20_kobj$ echo "123" | sudo tee mod_buff > /dev/null
vadim@vadim-RV520:/sys/kernel/hw20_kobj$ head -n1 mod_buff
123
vadim@vadim-RV520:/sys/kernel/hw20_kobj$ head -n1 /proc/hw20_mod
123
vadim@vadim-RV520:/sys/kernel/hw20_kobj$ head -n1 /dev/hw20_mod
123
vadim@vadim-RV520:/sys/kernel/hw20_kobj$
```

Выгружаем модуль, проверяем отсутствие файлов в **proc** и **sysfs**:

```
vadim@vadim-RV520:/sys/kernel/hw20_kobj$ cd ..
vadim@vadim-RV520:/sys/kernel$ sudo rmmod hw20_mod
vadim@vadim-RV520:/sys/kernel$ ls -ld hw20_kobj
ls: cannot access 'hw20_kobj': No such file or directory
vadim@vadim-RV520:/sys/kernel$ ls -ld /proc/hw20_mod
ls: cannot access '/proc/hw20_mod': No such file or directory
vadim@vadim-RV520:/sys/kernel$ cd /dev
vadim@vadim-RV520:/dev$ sudo rm hw20_mod
vadim@vadim-RV520:/dev$ dmesg | tail -n 10
[ 1007.413932] hw20_mod: read args: buff=*0x00000000093718d3e; buf_size = 8192.
[ 1013.924429] hw20_mod: Trying to read from buf.
[ 1013.924439] hw20_mod: read args: buff=*0x0000000008b8068b6; buf_size = 8192.
[ 1013.924479] hw20_mod: Trying to read from buf.
[ 1013.924482] hw20_mod: read args: buff=*0x0000000008b8068b6; buf_size = 8192.
[ 1117.482163] hw20_mod: Trying to read from buf.
[ 1117.482171] hw20_mod: read args: buff=*0x00000000fcde5a94; buf_size = 8192.
[ 1123.821015] hw20_mod: Trying to read from buf.
[ 1123.821024] hw20_mod: read args: buff=*0x000000008ec55476; buf_size = 8192.
[ 1188.990646] hw20_mod: Module unloaded.
vadim@vadim-RV520:/dev$
```